Related work: how does your lead do it?

Start from the lead you already chose (the paper or app). For each CORE item, answer:

1. What does it do? Does your lead have something like this?
2. How does it do it?
3. What's different for us? What will you follow, and what will you do differently?

If your lead doesn't cover a CORE item, find one other source that does and answer the same questions. Search with the item plus your product, for example "CNC homing limit switch" or "overhead crane emergency stop."

Lead Related Work

1. **Core: Routing/Movement**

What does it do? Does FarmBot have something like this?
Yes, FarmBot moves a tool head to specific coordinates within a defined workspace using X, Y and Z motion. This is similar to our gantry because our system also needs to know where the moving hook is and move it to a target location inside the workspace.

How does FarmBot do it?
FarmBot Genesis uses four NEMA 17 stepper motors. The motors have rotary encoders that provide closed-loop feedback about motor movement and position. FarmBot also supports homing/calibration and stall detection so the system can establish a known position and detect movement problems. Its drivetrain uses timing belts and pulleys for horizontal movement and a leadscrew for the Z axis.

What is different for us? What will we follow or do differently?
We can follow the same general ideas of starting from a known home position, tracking position and keeping movement inside defined limits. Our first software tests can use hard-coded coordinates before the physical gantry is ready. However, our system is an overhead material-handling gantry rather than a garden robot so the final motors, drivetrain and position-feedback method will need to account for the weight of the hook and payload. We also currently plan to focus mainly on X-Y gantry movement with lifting/lowering handled by the hook attachment.

2. **Core: Safety**

What does it do? Does FarmBot have something like this?
Partially, FarmBot has several safety-related features that are useful references for us. It uses encoder feedback and stall detection, supports homing, has defined movement limits and current Genesis hardware includes an emergency-stop control. These features help prevent uncontrolled or incorrect movement.

How does FarmBot do it?

FarmBot's stepper drivers include stall detection while rotary encoders provide closed-loop position feedback. The software can use known axis limits and homing/calibration to keep the machine operating within its intended area. FarmBot's electronics also include an E-stop control.

What is different for us? What will we follow or do differently?
We can follow FarmBot's use of position feedback, homing, movement limits and fault detection. Our safety requirements go further because our gantry will carry a suspended load. Our CORE safety feature also includes checking the load limit and stopping the system in an emergency. FarmBot is designed around lightweight garden tools so it does not fully answer how we should handle an overweight suspended load. For now, this is a difference and an area our team still needs to determine for our own design.

Supplemental related work for the missing load-safety part:
Konecranes' overhead crane safety guidance describes overload protectors that keep cranes from lifting beyond rated capacity and limit switches that slow or stop bridge/trolley travel near the ends of the allowed range. This is closer to our load-handling use case. For our prototype, we can follow the same principle to reject or stop an unsafe lift when the measured/known load exceeds our rated limit and use movement limits as another layer of protection.

3. **Core: Attachments**

What does it do? Does FarmBot have something like this?
Yes, FarmBot has an integrated universal tool mounting (UTM) that allows the machine to mount and remove different lightweight tools depending on the task. This is similar to our long-term idea of supporting different attachments even though our Version 0 will focus on one hook.

How does FarmBot do it?
The UTM uses magnetic coupling to hold tools and includes electrical connections plus liquid/gas ports so mounted tools can receive power, communicate with the system or use water/vacuum. FarmBot stores tools in tool bays and can automatically mount or dismount them.

What is different for us? What will we follow or do differently?
The main idea we can follow is creating a standard attachment connection so the gantry can support more tools later without redesigning the whole machine. For Version 0, we will keep the scope smaller and use one hook attachment that can lift and lower a load and work with our load-limit checks. FarmBot's tools are lightweight and task-specific while our hook will handle heavier suspended objects so our attachment needs stronger mechanical support and additional safety considerations.

Conclusion:
FarmBot is still a useful primary related work because it covers all three CORE areas at a general system level: controlled gantry movement, movement/position safety and interchangeable attachments. The largest gap is overhead load safety. For that part, an

overhead crane source such as Konecrances is a better reference because it directly addresses rated-capatcity overload protection and travel limit switches. FarmBot gives us useful ideas for position tracking, movement limits, fault detection and modular attachments but our team still needs to determine how those ideas should be adapted for a system carrying suspended loads.

References:

https://docs.farm.bot/
https://farm.bot/pages/genesis
https://farm.bot/pages/hardware
https://farm.bot/products/farmbot-genesis-v1-8

https://www.konecranes.com/sites/default/files/2019-08/Safety_White%20paper_Aug2019_Konecranes.pdf
https://portal.konecranes.com/amer/en/category/SP00005099/SP00005114